

IT214 Database Management Systems (Winter '2026)

Instructor: PM Jat, 2203, Faculty Block-2

Course Placement: B.Tech. (ICT), Semester-IV

Course format: 3 Lecture hours and 3 Lab hours.

Course content: This course is intended to give a solid background in database design and implementation with the focus on relational model. The course covers data modeling, theory of relational model, Querying relational databases using Relational Algebra and SQL. Theory of Functional Dependencies, Normal Forms, Decomposition Principles and Algorithms. Students also learn and practice creating stored procedures, accessing databases in host programming languages. The course also introduces Storage and Indexing structures, Query Execution and Transaction Processing concept.

During Lab hours, students practice concepts learned in theory and would be asked to design and implement a database system for a mid-size real life scenario.

Text Book:

1. **Fundamentals of Database Systems**, 5th ed., Ramez Elmasri and Shamkant B. Navathe, Pearson Education

Reference Books:

1. **Database Systems: The Complete Book** by Hector Garcia-Molina, Jeffery Ullman, and Janiffer Widom, Pearson Education
2. **Database System Concepts** by Avi Silberschatz, Henry F. Korth, S. Sudarshan, TMH
3. **Database Management Systems** by Raghu Ramakrishnan and Johannes Gehrke, TMH

Evaluation plan:

1. Three In Semester examinations: 30%
2. Database Project: 15%
3. Lab Evaluation: 15%
4. End Semester examination: 40%

Course Outcomes:

Database is one of core competency area for ICT undergraduate students. Through this course students develop ability to

- Design and implement Database systems.
- Write and solve complex queries systematically
- Write programs that access databases.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x	x						x			

Tentative Lecture Distribution

Sl. No.	Description		No. of Lectures
1	Introduction to Databases		3
	1.1	Database, Instance, Schema, and Metadata	1
	1.2	Conceptual and Implementation database Models	1
	1.3	3-Schema Architecture - Data and Operation Independence	1
2	Relational Model and Operations on Relations (All operations are discussed in Relational Algebra and SQL side by side)		8
	2.1	Relational Concepts	1
	2.2	Basic Relational Operations (RA and SQL)	2
	2.3	Set Operations (RA and SQL)	1
	2.4	Division Operation (RA and SQL)	1
	2.5	Aggregation Operations (RA and SQL)	1
	2.6	More SQL operations	1
	2.7	Examples (RA and SQL)	1
3	Entity Relationship Modeling		6
	3.1	Concepts and notations: Entity/Relationship type, instance, and sets	1
	3.2	Aggregation, Ternary Relationships and case study	1
	3.3	ER to relational Mapping	1
	3.4	Enhanced ER Modeling and EER to relational mapping	1
	3.5	Case study discussion	2
4	Database Design		7
	4.1	Database Design Concepts and Design Anomalies in relations	1
	4.2	Theory of Functional Dependencies	2
	4.3	Normal Forms as measure of “goodness” of a relation design	1
	4.4	Decomposition Principles and Algorithms	2
	4.5	Design Tutorials	1
5	Database Programming		5
	5.1	Views and Stored Procedures	2
	5.2	Triggers	1
	5.3	Embedded SQL	1
	5.4	API based access of databases	1
6	Indexes		4
	6.1	Database storage file structures	2
	6.2	Indexing Methods and SQL support	2
7	Transaction Processing		5
	7.1	Concepts	2
	7.2	SQL Isolation Levels	1
	7.3	Concurrency Control Techniques	2
8	Query Execution and Optimization		4
	7.1	Query Execution and Optimization Concepts	2
	7.2	Algorithms for basic operations	2
Total Lectures			42